

No Observed Difference in Zooplankton Species Presence at Different Locations in Lake Matoaka

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Abstract

Lourens Baas Becking established the idea that microorganisms are able to disperse everywhere, and the environment selects who persists. In Lake Matoaka, an artificial, hypereutrophic lake by the campus of William & Mary, there are no substantial barriers to dispersal of aquatic microorganisms. However, the lake receives inputs from a variety of streams with differing water quality. In this project, five locations in Lake Matoaka with different water inputs were sampled and observed for species presence to investigate whether zooplankton are all able to both disperse to and persist in those local environments. We collected water samples with a plankton net and examined the samples under microscopes for species identification. Major zooplankton groups identified include Copepoda, Rotifera, and Cladocera. Overall, all three major zooplankton groups were found at every sample site. This suggests that common zooplankton species are widely dispersed and persist in Lake Matoaka despite differences in local environment.

Keywords: Zooplankton, Hypereutrophic, Lentic, Dispersal, Selection

1 Introduction

1.1 “Everything is everywhere, but the environment selects”

Plankton are important components in freshwater ecosystems. They serve as a basal component of the aquatic food web, promoting nutrient cycling and energy transfer. They include a wide range of taxa, such as rotifers, *Daphnia*, copepods, and larvae of many aquatic invertebrates. Their distributions are influenced by water conditions including chemistry, oxygen content, and nutrient availability. In addition, biotic interactions, like fish predation and grazing on phytoplankton, alter their distribution and abundance. As a result, zooplankton is an important part of understanding how environmental changes influence aquatic ecosystems.

In Baas Becking’s 1934 *Geobiologie of inleiding tot de milieukunde*, the hypothesis that every microorganism is able to get everywhere, but the local environmental conditions select which species occupy an area was popularized [1]. Microorganisms have large populations, small sizes conducive to a variety of dispersal methods, and sometimes tolerant traits that allow harsh journeys or destination environments. For example, *Daphnia* produce resting eggs enclosed in ephippia, allowing them to survive in sediments

for years and regenerate in favorable environments after being transported between water bodies. These characteristics allow dispersal of organisms “everywhere”. Within Lake Matoaka, it is likely that plankton can move around relatively freely in most cases, given the mild currents and lack of extreme barriers.

The variety of water inputs with different conditions connected to the lake lead to diverse local environments within the lake. These inputs may vary in nutrient concentrations, oxygen content, and the degree of urban or natural influences. This might create unique microenvironments near the water inputs that support different plankton communities. In this project, we investigate whether these differing conditions lead to different zooplankton species compositions.

1.2 Lake Matoaka

Lake Matoaka is a hypereutrophic, shallow, man-made, 16-hectare lake induced at William & Mary’s campus in Williamsburg, Virginia as a mill pond in the early half of the 1700’s. Increasing development of campus and urban areas over the years contributes to lake eutrophication [2]. This lake has a history of sewage spills, excess algal blooms, and closure to public human recreation for safety and liability concerns [3]. Lake Matoaka is used recreationally during warmer months by the college’s community and is home to a variety of native and invasive flora and fauna. The Eastern side of the lake borders the campus of William & Mary, while the Western side is surrounded by a more expansive buffer of wooded area. Lake Matoaka takes water inputs primarily from four streams, with unique input conditions that originate from highly developed to primarily wooded areas. For example, downstream areas with runoff from more developed areas are found to have higher nutrient concentration, like phosphorus, compared to runoff from the forests around [2]. The water flows out from Lake Matoaka into College Creek via the dam under Jamestown Road.

We chose four input locations and one output location around the lake to sample from, representing the water conditions of various influents, and the mixing of all inputs before water exits to College Creek under Jamestown Road.

1.3 Research Questions

Our project aims to answer the following questions: (1) Is the zooplankton composition the same across different locations in Lake Matoaka, which have stronger influences from different water inputs? and (2) What plankton exist in different areas of Lake Matoaka?. Initially, there were plans to consider phytoplankton, so photographs were collected, but identifications remain incomplete due to time constraints.

We hypothesize that zooplankton composition would differ among the five sample sites due to varying influent conditions. We predicted that different water qualities, such as increased pollution and nutrient content on certain inputs would lead to different, and perhaps more/less tolerant, taxa occupying areas of the lake most local to those inputs. Our null hypothesis is that the zooplankton composition is similar among the five sample sites. Our alternative hypothesis is that the zooplankton composition is different among the five sample sites.

2 Methods

2.1 Sampling Sites

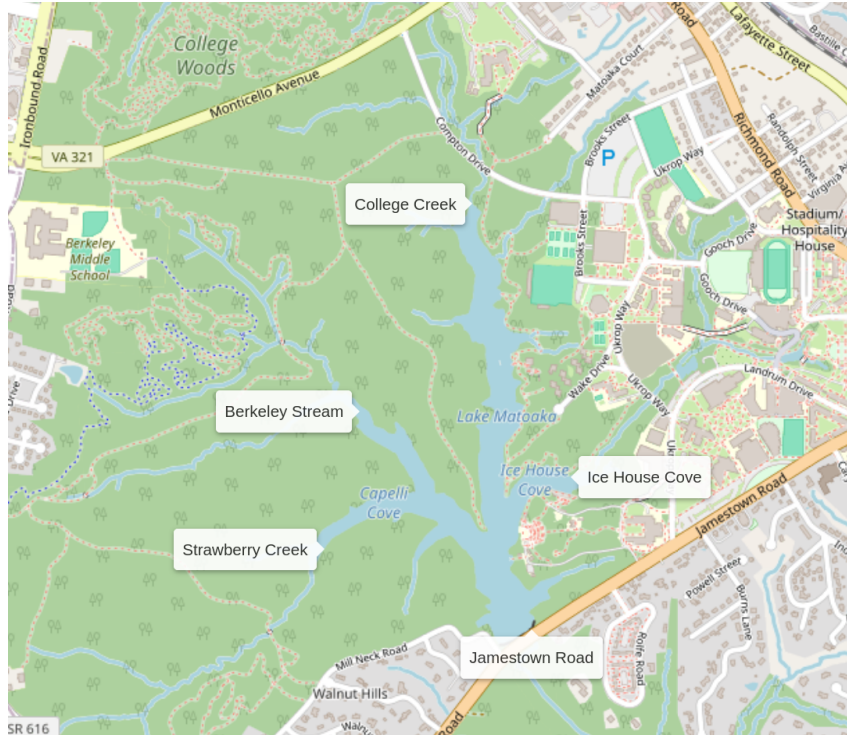
We took samples at five sites in Lake Matoaka (Table 1), representing four different sources of water inputs and one output location, hypothetically combining the inputs from all the sources.

2.1.1 College Creek

College Creek inputs water to Lake Matoaka from its branching streams in the north of campus, where the College Woods borders highly developed residential and commercial areas. On the North and West

Table 1 Sampling locations at Lake Matoaka

Location Name	Coordinates of Sample	Timestamp of Sample
College Creek	37.27455°N, 76.72408°W	1:30 PM, 24 March 2026
Ice House Cove	37.26761°N, 76.72095°W	1:50 PM, 24 March 2026
Jamestown Road	37.26373°N, 76.72219°W	2:02 PM, 24 March 2026
Strawberry Creek	37.26574°N, 76.72881°W	2:17 PM, 24 March 2026
Berkeley Stream	37.26928°N, 76.72772°W	2:32 PM, 24 March 2026

**Fig. 1** Map of sampling sites

sides, it receives runoff from the Commons dining hall/Kaplan arena area of campus, and commercial and residential areas between Ironbound Road, Monticello Avenue, and Richmond Road. In this paper, we refer to the part of College Creek described here when speaking of sampling locations, not to be confused with the much larger College Creek downstream of Lake Matoaka, which acts as a tributary to the James River. As a result, input from College Creek would contain a higher level of nutrients compared to other sample sites.

2.1.2 Ice House Cove

Ice House Cove brings inputs from Tinkling Rill, a shallow lotic system connecting the Swem Dell, Grim Dell, and Crim Dell ponds. It collects runoff from the center of campus, surrounded by roads, dining halls, sports fields, dorms, and academic/administrative activity. This makes Ice House Cove an accumulation point for campus-derived materials, which then flow into Lake Matoaka.

2.1.3 Strawberry Creek

Strawberry Creek receives runoff primarily from the trails in the College Woods (on its creeks), and it stems from a stormwater retention pond between John Tyler Highway and Humelsine Parkway, which receives runoff from the surrounding roads and developed area. The water input from Strawberry Creek is influenced by both natural and anthropic factors.

2.1.4 Berkeley Stream

Berkeley Stream branches throughout the Matoaka Trails in the College Woods, forming a network of small channels in the forest. It receives runoff primarily from this wooded area, which is mainly affected by the soil, leaf litters, and vegetation along the stream. Although the far ends of it reach the nearby Berkeley Middle School and a housing development, the water input from Berkeley Stream is least affected by human activities among our five sample sites.

2.1.5 Jamestown Road

Jamestown Road is the name we gave the area where Lake Matoaka meets the Matoaka Dam beneath Jamestown Road. It becomes sandier and shallower and is downstream of all influents other than runoff from Jamestown Road itself. The specific area we sampled is close to the tunnel through the dam where water can pass through to College Creek.

2.2 Sampling

All sampling took place within about an hour in the early afternoon of March 24, 2026. At the time of sampling, the annual large algal blooms and pollen deposits had just begun to establish, and the air temperature was about 49 degrees Fahrenheit. Location coordinates were recorded by taking the current latitude and longitude from Apple Maps during sampling.

The community composition found here is most relevant to the early Spring time frame due to seasonal population changes, when the water body starts to mix. The plankton population usually increases due to rising temperature and enhanced nutrient cycle. It is also important to note that if additional sampling days were needed, they would have had to occur at the same time of day to account for diel vertical migration.

Samples were collected by canoeing out to each sampling site, and filling one small glass bottle about two-thirds full per site (Figure 2). Only a couple drops of water would be inspected per site, so the initial sampling volume was approximated to be this sufficient volume. To do this, we threw out a plankton net (Figure 3) three times per deposit of water into our bottle, needing one to two deposits to collect enough water.

The areas sampled were shallow enough that the entire depth at the location could plausibly be considered the epilimnion due to high sunlight influence. We sampled as far into each source stream as possible while avoiding the beginning of the floating Spring algal overgrowths and staying at an area deep enough to throw the plankton net. Samples were preserved in ethanol and stored in the fridge. We photographed and observed plankton for six microscope slides per sample.

2.3 Imaging

Photographs were taken with phone cameras through a table-top microscope optic, primarily at 100x and 400x magnification. Due to time constraints and a focus on the presence aspect of community makeup, species abundance and organism sizes were not measured.

2.4 Identification

A variety of online sources were used to identify plankton. For zooplankton the primary reference was the University of North Hampshire's "Image-Based Key To The Zooplankton Of North America" [4]. For



Fig. 2 Samples before preservation in ethanol



Fig. 3 Plankton net used to collect samples

phytoplankton, the Fluid Imaging guide [5] was used in general, the diatoms.org key [6] was used for diatoms, and the National Institute of Water and Atmospheric Research in New Zealand desmid quick-guide [7] was used for desmids. Some additional resources referenced are the “Key for the identification of crustacean nauplii” by Dahms, et al [8], the “Practical Guide to Identifying Freshwater Crustacean Zooplankton” from Laurentian University [9], the University of California Santa Cruz Phytoplankton Identification page [10], copepedia.org’s plankton database [11], and algaebase.org’s algae database [12].

3 Results

We observed three primary zooplankton (Figure 4) and a variety of phytoplankton across all sampling sites. The zooplankton consisted of rotifers, cladocerans, copepods, and copepod nauplii. We found some ambiguous nauplii stages, which we attributed to be very young copepod nauplii. In addition, we also observed some folded and disfigured organisms, which were difficult to give precise identification due to limited morphological features after folding.

Table 2 Observation of zooplankton existence at each sampling site.

Site	Was this zooplankton observed?		
	Copepoda	Rotifera	Cladocera
Berkeley	Yes	Yes	Yes
College Creek	Yes	Yes	Yes
Ice House Cove	Yes	Yes	Yes
Jamestown Road	Yes	Yes	Yes
Strawberry Creek	Yes	Yes	Yes

Overall, the same groups of zooplankton were observed at every sampling site (Table 2). It might indicate an even distribution of zooplankton within Lake Matoaka. This supports the idea that zooplankton are able to move around Lake Matoaka freely, allowing them to be everywhere and selected by the environment. It also reflects the lack of physical barriers in Lake Matoaka, which allow the zooplankton to disperse.

The rotifers observed appear to be dominant by *Keratella cochlearis*, which is commonly found in freshwater. The majority of copepods observed appear to belong to the order Cyclopoida, with a small portion of order Calanoida. The cladoceran population seemed to consist primarily of Bosminidae.



Fig. 4 Example images of the main zooplankton taxa observed. (From left to right) Top row: Cyclopoida sp., Calanoida sp., Copepod nauplius; Bottom row: Bosmina, Keratella cochlearis, unknown creature assumed to be a copepod nauplius

4 Discussion

As we observed rotifers, copepods, and cladocerans, it seems no particular environment sampled had conditions non-conducive to occupation of any of these three prominent zooplankton groups. This does not support our hypothesis that different species of zooplankton would be found at different sample sites. It should be noted that we only examined the composition of the zooplankton, not the phytoplankton. However, it supports the first half of Baas Becking’s hypothesis that everything is everywhere.

It is possible that the locations and inputs around Lake Matoaka were so similar that the environmental selection taking place cultivated similar ecosystems. It is also possible that the environmental difference is not strong enough to fully exclude a certain species. A previous study by Pensa and Chambers (2004) shows that the downstream areas of the more developed subwatershed had higher levels of phosphorus and biogenic silica, indicating a greater input of nutrients into Lake Matoaka [2]. Future study should measure water quality information, including nitrogen content, phosphorus content, and oxygen saturation, to determine whether water conditions differ among sample sites. It should also examine the correlation between water quality and plankton composition, which allow us to test for the second half of Baas Becking’s hypothesis that the environment selects.

In addition, plankton are able to disperse so fast in this lake that the organism presence from dispersal may overwhelm the effect of selection. Seasonal succession might also contribute to zooplankton distribution. In spring, water columns are partially mixed due to change in water density. The mixing process could redistribute the zooplankton with water current, which causes the zooplankton to appear at every sample site.

The abundance, genus, and size of rotifers and copepods, although not quantified in this study, aligned with the correlation drawn between communities being composed primarily of small-bodied rotifers and cyclopoid copepods and lakes being hypereutrophic by Karpowicz, et al (2025) [13].

The data collected only accounts for boolean presence. Species dominance may vary across locations, indicating some environmental selection favoring certain species at certain locations’ conditions. We were unable to collect abundance data due to time constraints. However, there are noticeable differences in species in the sample we observed. For example, Strawberry creek shows a higher abundance of copepods and cladocerans compared to rotifers. Future work including this may align with or challenge previous findings that although species may have high dispersal rates and the ability to occupy diverse sets of environmental conditions, different local conditions cultivate species dominance and community makeup [14]. When considering abundance, it may also be important to take carcasses into account, avoiding a distorted community composition from counting dead bodies. Methods such as the use of aniline blue for living/dead distinctions [15] may be incorporated to allow this.

Although we saw no difference in zooplankton composition across sites, we suspect there may be differences in phytoplankton composition due to their limited mobility. Given more time, we could identify more of the photographed phytoplankton and decisively claim this. However, by visual inspection, we did observe some varieties of diatoms only at some sites, some desmids at only some sites, etc. Future work could identify all the observed phytoplanktons and compare their species and abundance between sample sites. Additionally, previous study by Li et al. (2019) found a synchronized relationship between zooplankton and phytoplankton when responding to changes in environmental factors [16]. Future work should also combine the data to investigate whether there is a relationship between zooplankton and phytoplankton composition that aligns with the previous findings.

Due to the small sample size (one sample per site, six slides per sample), no statistical claims about the observations here can be made. Future work should include sufficiently large sample sizes to make such claims.

Given the time constraint, not every plankton photographed has been identified yet, especially for phytoplanktons. Some further identifications than mentioned here, as well as the full gallery of identified images can be found on this project’s website, hosted at <https://gummyhz.github.io/matoaka-plankton/>. All photographs of observed organisms (whether identified or not) are available in the assets folder of <https://github.com/gummyhz/matoaka-plankton>.

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